STROBE reporting checklist

Adapted cross-sectional checklist for an ecological national registry time series. STROBE is a reporting aid, not evidence of design validity or author approval. Locations refer to section headings, which remain stable across Word/PDF pagination.

Source. https://www.strobe-statement.org/checklists/

Item	Reporting topic	Location or applicability
1a–b	Design and balanced abstract	Abstract explicitly identifies ecological design and distinguishes national uncertainty from young-domain findings.
2	Rationale	Introduction, paragraphs 1–3.
3	Objectives	Introduction, final paragraph; sensitivity analyses explicitly post hoc.
4	Design	Methods, Study design and reference reproduction.
5	Setting and dates	Brazil, 2014–2021; Methods and supplementary provenance.
6a	Eligibility and selection	Methods, Selection and Calendar ages; Supplementary Table S1. Population records rather than recruited participants.
7	Variables	Underlying-cause J45/J46, age domains, calendar year and mortality rates defined in Methods; denominator and standard weights specified.
8	Sources and measurement	SIM, IBGE revisions, date algorithm and public dictionary; Methods and supplementary provenance.
9	Bias	Selection sensitivities, ambiguity of duplicates and denominator revisions; Methods and Discussion limitations.
10	Study size	All eligible national underlying-cause records within the period; no sampled recruitment or prospective power calculation.
11	Quantitative variables	Completed calendar age and single-year/90+ bins; nine count-model bands and continuous year covariate in Methods.
12a	Statistical methods	Direct standardisation, rate GLMs/OLS, count families, offsets and intervals; Methods.
12b	Subgroups	5–34, nested 7–34, 35–59 and ≥60; Methods, Results and Supplementary Tables S2, S5, S7, S8. No interaction testing reported.
12c	Missing data	Invalid/missing dates excluded and separately logged; missing count cells assigned zero and positive complete populations required. Table S1.
12d	Sampling strategy	Not applicable to this national registry enumeration; no sampling weights. Standardisation weights are defined separately.

Item	Reporting topic	Location or applicability
12e	Sensitivities	One-condition selection and period scenarios, both revisions/standards and all count families; Methods and S4–S8.
13a–c	Flow and exclusions	Results, Historical reproduction and corrected selection; Table S1 and preserved selection logs. No individual recruitment or follow-up.
14a	Descriptive characteristics	Age-specific national counts and rates in S2. Analyses pool sexes; no sex-specific associations are claimed.
14b	Missingness	Table S1 reports exclusions for invalid dates/negative ages. Other registry-field missingness is not used as an exclusion criterion.
15	Outcome data	Annual deaths and aligned denominators in S2 and full-precision CSV outputs.
16a–c	Main results	Table 1, Results and S4–S8 show estimates, intervals and endpoint changes; adjustment/age-band rules described in Methods. Absolute rate change is reported, not individual risk.
17	Other analyses	Kitagawa six endpoint pairs and prespecified execution scenarios; post hoc status acknowledged; S6–S8.
18	Key results	Discussion opening and young-domain interpretation match study objectives.
19	Limitations	Ecological design, short series, certification, record ambiguity, historical hashes, multiple comparisons and theta warnings; Discussion.
20	Interpretation	Discussion distinguishes trend uncertainty, endpoints and descriptive decomposition; no causal attribution.
21	Generalisability	National Brazilian registry analysis; ecological results do not establish individual associations or historical reversal.
22	Funding	Title page states no funding was received, as confirmed during revision preparation. No sponsor role is reported.